

L^AT_EX in the Digital Humanities

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Abstract

This paper explores the intersection of L^AT_EX typesetting and the digital humanities. More specifically, it asks why L^AT_EX typesetting is used so infrequently in the Digital Humanities (DH), despite its clear advantages and applications. This paper, on the one hand, aims to understand the reasons why L^AT_EX isn't more widely used in the digital humanities and, on the other hand, presents three examples of relevant use cases to illustrate its value for the (Digital) Humanities.

1 What are the Digital Humanities?

Despite having become established as a field in recent decades, defining Digital Humanities (DH) remains difficult because the field encompasses a range of methods, tools, and subject materials. It integrates both digital and computational approaches with Humanities research as an interdisciplinary field.¹ Broadly speaking, Digital Humanities refers to the use of digital methods to analyze Humanities data and to address Humanities research questions. Roth [13] has proposed distinguishing three subfields:

1. **Humanities of the Digital:** This branch focuses on media studies and similar matters.²
2. **Digitized Humanities:** As suggested by the name, this means hermeneutically traditional humanities work (such as text editing, annotation, or interpretation) but moved to digital spaces and technologies such as TEI-XML (to be discussed later).
3. **Computational Humanities:** Here, advanced computational techniques—such as machine learning, large-scale text analysis, or computer vision—are applied to Humanities data. While these methods offer powerful new approaches, they often come with unexpected challenges in applying them to humanities data. This often proves surprisingly difficult and the methods follow epistemologies fundamentally different from traditional Humanities work.

2 Use cases for L^AT_EX in DH

1. **Using L^AT_EX in conference submissions** is a more universal approach to the use of L^AT_EX,

primarily related to the presentation and publication of research rather than the content itself. Its benefits lie in how effectively it handles complex documents, making it a useful tool across various fields, including the humanities.

2. **For archaeological and other catalogues or large scholarly monographs** in the humanities, L^AT_EX is very well suited. Monographs replete with images tend to be unwieldy in software like MS Word, whereas L^AT_EX handles these with ease and isn't that hard to learn, even for humanists.³
3. **In digital scholarly editions**, producing print versions is probably one of the most interesting uses of L^AT_EX for Digital Humanists [10, 11].

We'll discuss each of these in more detail.

2.1 Use case 1: Conference proceedings in L^AT_EX

Conference proceedings that require full paper submissions, such as those for by the *Computational Humanities Research* Conference present a good reason to get acquainted with L^AT_EX.⁴ However, the use of L^AT_EX has been criticized in some Digital Humanities (DH) circles as exclusionary, as noted by Dombrowski [1]. While I agree that not all DH work requires coding, I don't see why L^AT_EX is particularly singled out, especially since it is not as difficult to learn compared to other tools increasingly common in DH. The sentiment of alienation seems less about the technology itself and more tied to the association of L^AT_EX with Computer Science, a field not known for pioneering inclusivity and diversity.

In my view, there are several advantages to using L^AT_EX for conference submissions: It allows the outsourcing of typesetting work to authors via standardized L^AT_EX templates. This makes sense for smaller conferences run by a community, though it does place some additional burden on editors and organizers. Humanities scholars, even those with limited technical expertise, can learn L^AT_EX as a markup language to format their submissions effectively. Many are already familiar with markup languages, making L^AT_EX far easier to grasp than programming languages like Python, which are becoming more prevalent in DH.⁵ Additionally, L^AT_EX facilitates fast, open-access publication of conference proceedings, which benefits early-career scholars and provides an alternative to the exploitative practices often seen in commercial academic publishing.

¹ Some even argue that field doesn't want to define itself [12]. My definitions follow [6], [12], and [13].

² This is not my flavour of DH and seems to me especially dominant in the Anglophone area.

³ See an example of typesetting archaeological catalogues in [7].

⁴ computational-humanities-research.org

⁵ See tutorials specifically for Humanists in [4, 8].

2.2 Use case 2: Printed books and archaeological catalogues

2.2.1 Physical books in the Humanities

Despite early predictions that digital media would render physical books obsolete, by 2024 it is evident that this shift has not occurred and seems unlikely to happen in the near future. The value of the book as a material object and a symbol of cultural capital has retained its significance. It has even gained renewed symbolic value among younger audiences on platforms like BookTok or celebrities who hire book curators for their homes and to decide which books they should be photographed holding. This suggests a renewed appreciation of books not only as material objects but also as symbols of cultural capital. Moreover, many scholars in the Humanities continue to prefer printed materials for tasks such as close reading, underscoring the book's enduring role in academic practice.

2.2.2 L^AT_EX in archaeology

In disciplines such as archaeology, compiling extensive catalogs of objects or findings is a common component of larger research endeavors, such as PhD dissertations. These catalogs often contain numerous images and, due to their size, can become difficult to manage using standard word processing tools like MS Word. While this usage extends beyond the Digital Humanities, it highlights the cross-disciplinary relevance of L^AT_EX.

As previously discussed, L^AT_EX excels in typesetting large scholarly monographs and managing extensive catalogs. Its capacity to handle large datasets (e.g., with tools like `csvsimple`, see [7]) and incorporate numerous images efficiently makes it an ideal tool for managing complex archaeological projects.

2.3 Use case 3: Print versions for digital scholarly editions

Perhaps the most significant use case for L^AT_EX within the Digital Humanities is digital scholarly editing (DSE [14]). DSE is a core task in the DH community. This is particularly true in contexts such as my institution in Graz, where DSE traditionally holds a key position, even as the subfield of Computational Humanities gains momentum. This leads me to assume that digital scholarly editing will likely remain a core technology within Digital Humanities.

Given this, and as physical books and print editions persist and show no signs of disappearing, L^AT_EX remains relevant for the production and dissemination of scholarly editions and research outputs.

2.3.1 Transforming TEI-XML to L^AT_EX

The Text Encoding Initiative (TEI) is the standard XML format used for archiving and managing data in digital scholarly editions.

Transforming digital scholarly editions in TEI-XML format to L^AT_EX enables the creation of high-quality print versions from digital editions, even when the primary focus of the edition is not on print publication. Using XSLT to transform TEI-XML data into L^AT_EX facilitates the generation of print-ready material with minimal effort. This process could potentially be enhanced by the use of large language models to automate the generation of XSLT code (or similar), further streamlining the workflow for scholars in the future. This caters to the continuing demand from traditional humanists for print(-able) editions. These print versions are particularly beneficial in educational settings where physical materials remain a preferred medium for study and instruction.

The `reledmac` package [16] is especially valuable for Digital Humanists to ensure the accessibility and usability of scholarly resources in both digital and print formats, as it supports complex typesetting requirements often encountered in scholarly editions [2].

2.3.2 TEI primer

The following is a minimal example of TEI-XML:

```
<TEI> <!-- root element -->
<teiHeader>
<!-- author, title, dating,
        sources, edition rules, etc. -->
</teiHeader>
<text> ... </text>
</TEI>
```

TEI further offers elements such as `<div>` (division) for a text division, `<hi>` (highlight), `<pb>` (page beginning) and a range of specialized modules. Thus, it allows for the encoding of complex textual information, such as multiple versions of a text or critical apparatus for documenting textual variants across different witnesses in a stemma.

In TEI, a critical apparatus is encoded using several elements to capture textual variants across witnesses. The `<app>` (apparatus entry) containing one entry in a critical apparatus,⁶ with `<rdg>` (reading) and `<lem>` (lemma) tags capturing the actual variations. Multiple readings can be grouped within a `<rdgGrp>` if necessary.

Notably, `<lem>` can contain text that does not appear in any of the textual witnesses, allowing for

⁶ tei-c.org/release/doc/tei-p5-doc/en/html/TC.html#TCAPLL

editorial interventions or reconstructions of the text in the base text. The following shows an example:

```
<app>
  <rdgGrp type="orthographic">
    <rdg wit="#Sh1">giue</rdg>
    <lem wit="#Sh2">give</rdg>
  </rdgGrp>
  <rdg wit="#AS1">have</rdg>
</app>
<listWit>
  <witness xml:id="Sh1">
    <bibl>Folger STC 22276</bibl>
  </witness>
  <witness xml:id="Sh2">
    <bibl>Huntington 69304</bibl>
  </witness>
</listWit>
```

2.3.3 Typesetting critical editions with the `reledmac` package

A critical apparatus documenting textual variants found in different witnesses of a text's transmission requires specialized tools for typesetting; the `reledmac` package is designed to manage such complex editorial needs. TEI encodes text semantically using XML, but to produce a print version, it needs to be transformed into the presentational markup of L^AT_EX.

The TEI Apparatus Toolbox (TEI-CAT), developed by Marjorie Burghart, offers a ready-made solution for this transformation.⁷ It allows scholars to print an edition of a TEI-XML document by converting it into L^AT_EX using the `reledmac` package and subsequently compile it to PDF. This transformation is achieved via an XSLT script, customizable through a web interface. The tool offers a practical way to automate the typesetting of TEI-encoded critical editions, although it does have limitations in terms of fine-tuning the final layout and appearance.⁸

The XSLT template used for transforming TEI into L^AT_EX using `reledmac` is quite complicated, as it has to account for various encoding possibilities within TEI and integrates the customization options from the web form. However, this is hard to follow for experts in XSLT and far too difficult for beginners. Thus, I developed simplified XSLT templates for teaching purposes, aimed at helping learn-

ers understand the basics of transforming TEI into L^AT_EX [5].⁹

3 Outlook

3.1 Popularizing L^AT_EX in Digital Humanities

While L^AT_EX has significant potential in the Digital Humanities (DH), its adoption remains limited. In DH, many projects prioritize digital outputs over printed formats. Despite the clear advantages of L^AT_EX for typesetting and managing large, complex documents, it remains underutilized in the field.

However, XSLT-based transformations offer a practical way to generate high-quality print versions from TEI-encoded digital editions, a crucial capability as the alternative XSL-FO (Formatting Objects) standard was discontinued in 2012. Additionally, the rise of large language models presents new possibilities for automating and simplifying these transformations, making the creation of print-ready editions from TEI data even more accessible.

Print outputs remain valuable, particularly for textual scholars who continue to favor working with printed materials for tasks such as close reading and annotation. By addressing these opportunities, L^AT_EX could become more integral to the DH toolkit.

3.2 Practical implementation challenges in the GAMS repository

In the course of working on a digital scholarly edition project, I encountered several practical challenges with using L^AT_EX for generating print outputs intended for teaching materials [3].

Our editions are archived in the Humanities Asset Management System (GAMS) [15], which holds XML data from which it dynamically generates various representations such as HTML, print or RDF, following the single-source principle. This included creating PDFs from L^AT_EX through a multi-step process. The archived XML data used to be transformed into L^AT_EX code using XSLT via the `getLaTeXPDF()` method, and the L^AT_EX was then compiled into PDFs

⁷ `teicat.huma-num.fr/print.php`

⁸ For more details on the XSLT used in this process, see the XSL file, available here: github.com/MarjorieBurghart/TEI-CAT/blob/master/tei2latex_final.xslt. The transformation is triggered by a web form through which limited customization is possible.

⁹ These simplified XSLT documents were a key component of the first (and, thus far, last) in-person L^AT_EX Ninja workshop, held in 2022 at Harvard University [9].

The workshop provided participants with hands-on experience in using these templates to convert TEI into L^AT_EX, making the learning curve for beginners more manageable while still covering essential concepts of the transformation process. This included a simplified version of the TEI-CAT template used to generate `reledmac` L^AT_EX from TEI-XML, which is available here: github.com/sarahalang/Harvard_BeyondTEI_Workshop_SLang2022/blob/main/ADDITIONAL_RESOURCES/XSL_BASE_STYLESHEETS/mini-latex-reledmac.xsl

using Python’s Rubber.¹⁰ However, this process encountered several issues.

In the case of the transformation to L^AT_EX, the underlying L^AT_EX code was not always visible, making errors cryptic and time-consuming to resolve. Compounding this, even documents that had been successfully compiled in the past sometimes produced errors in subsequent attempts, adding unpredictability to the process. To troubleshoot, I often had to run the XSLT stylesheet locally, contrary to the intended automatic transformation on the system. This step added extra work, as it became necessary to manually verify the outputs.

A subsequent “dockerization” of GAMS complicated matters further, as some L^AT_EX functionalities were not carried over due to their limited use. This affected the `getLaTeX()` function that used to show the L^AT_EX code generated through the XSLT transformation. Without the ability to inspect the generated L^AT_EX code before compilation, debugging became even more difficult. Errors often stemmed from the XSLT transformation and resulted from discrepancies between XML and L^AT_EX, such as nesting and white-space handling. Despite efforts to include BibT_EX in the L^AT_EX file using `filecontents`, the system failed to compile it properly. The container used for compiling L^AT_EX was sizeable and retained outdated auxiliary files, often requiring manual intervention.

Ultimately, dynamic L^AT_EX generation from XML proved overly complex, resource-intensive, and prone to errors. Most users needed static outputs rather than dynamic ones, making it more efficient to compile L^AT_EX locally and archive static PDFs. This was already evident in the aforementioned GRAF project, where attempts to dynamically generate L^AT_EX outputs for teaching materials proved cumbersome for both stylesheet developers and system administrators. Ultimately, the team opted for manual compilation and static PDF archiving as a simpler, more reliable solution and discontinued the dynamic compilation from XML.¹¹ Moving forward, it will be necessary to compile L^AT_EX locally and upload static outputs archived in the repository.

Still, XML-to-L^AT_EX transformations may remain useful tools for complex print outputs, such as critical editions, in the digital scholarly editing community, despite the fact that these use cases were rare in the Graz GAMS infrastructure and thus, difficult to maintain.

3.3 Future directions

A tutorial and starter XSLT documents could significantly lower the barrier to entry for new users, providing a practical foundation for those unfamiliar with these workflows. I am considering creating a tutorial on the *Programming Historian*, teaching L^AT_EX basics and TEI-to-L^AT_EX via XSLT transformations in an accessible way, as well as explaining their value for digital humanists.¹² This step-by-step guide would demonstrate the process of converting TEI-encoded texts into L^AT_EX using XSLT and walk users through the creation of print-ready PDFs from XML-encoded digital editions.

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¹⁰ pypi.org/project/latex-rubber/

¹¹ A final documentation of the removed features can be found here: github.com/sarahalang/latex-for-dh/blob/main/latex-on-gams.md

¹² programminghistorian.org

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